

Non-canonical functions of cell cycle cyclins and cyclin-dependent kinases

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Abstract | The roles of cyclins and their catalytic partners, the cyclin-dependent kinases (CDKs), as core components of the machinery that drives cell cycle progression are well established. Increasing evidence indicates that mammalian cyclins and CDKs also carry out important functions in other cellular processes, such as transcription, DNA damage repair, control of cell death, differentiation, the immune response and metabolism. Some of these non-canonical functions are performed by cyclins or CDKs, independently of their respective cell cycle partners, suggesting that there was a substantial divergence in the functions of these proteins during evolution.

Progression through the cell division cycle is driven by cyclins, which bind to and activate their catalytic partners, the cyclin-dependent kinases (CDKs). Specific heterodimeric cyclin-CDK complexes phosphorylate a plethora of cellular proteins to promote entry into and progression through the G1 phase of the cell cycle, to drive DNA synthesis (during S phase) and to trigger segregation of the newly duplicated chromosomes to the daughter cells during mitosis, thereby ensuring that the cell cycle progresses in an ordered manner^{1,2}.

Mammalian cyclin and CDK families each contain more than 20 members³, but only a few cyclin-CDK complexes are known to directly participate in the cell division cycle (FIG. 1). Growth factors induce the expression of D-type cyclins (cyclins D1, D2 and D3), which are therefore regarded as molecular links between the cell environment and the core cell cycle machinery. Once induced, D-type cyclins interact with CDK4 or CDK6 and phosphorylate the pocket proteins pRB, p107 and p130, which bind to and regulate E2F transcription factors during the G1 phase of the cell cycle. Later during G1, E-type cyclins (cyclins E1 and E2) become upregulated and activate CDK2 (and, to a lesser extent, CDK1 and CDK3), resulting in phosphorylation of a broader range of cell cycle-related proteins. The subsequent induction in S phase of cyclin A2, which partners with CDK2 and CDK1, and the activation of cyclin B1-CDK1 at the onset of mitosis, drive the progression of cells through the remainder of the cell cycle through the phosphorylation of a large number of proteins that are involved in DNA replication, as well as in centrosome and chromosome function^{1,2,4,5}.

Consistent with a role for cyclin-CDK complexes in promoting cell cycle progression, amplification, genetic rearrangements and/or overexpression of cyclin or

CDK genes have been documented in nearly all human tumour types^{6–9}. In particular, the cyclin D1 gene (*CCND1*) is the second most frequently amplified locus across all human cancers. For these reasons, cyclin-CDK complexes are considered to be promising targets for cancer therapy.

Although the canonical role of cyclins and CDKs as essential drivers of cell cycle entry and progression has been firmly established (FIG. 1), research carried out over the past two decades provides increasing evidence for additional functions of these proteins in gene transcription, DNA damage repair, cell death, cell differentiation and metabolism. In this Review, we focus on the main biological areas in which such non-canonical functions of cyclins and CDKs have been documented. We restrict our Review to interphase cyclins and CDKs — that is, those operating in the G1, S and G2 phases. Non-cell cycle CDKs, such as CDK5, are only discussed in the context of their interactions with interphase cyclins, such as cyclin E-CDK5 function in post-mitotic neurons.

Kinase-dependent roles in transcription

In the current model of cell cycle regulation, the main function of cyclin D-CDK4 and cyclin D-CDK6 complexes is to phosphorylate and inactivate pocket proteins, thereby enabling the transcription of genes that are required for cell cycle progression^{2,5,10}. In addition, cyclin D-CDK complexes have been shown to phosphorylate several transcription factors, thereby linking the cell cycle and transcription (TABLE 1). Cyclin D-CDK4 complexes phosphorylate SMAD3, which inhibits the anti-proliferative effect of signalling through the transforming growth factor- β pathway¹¹ (FIG. 1). Cyclin D-CDK4 and cyclin D-CDK6 complexes activate the transcription factor forkhead box protein M1 (FOXO1),

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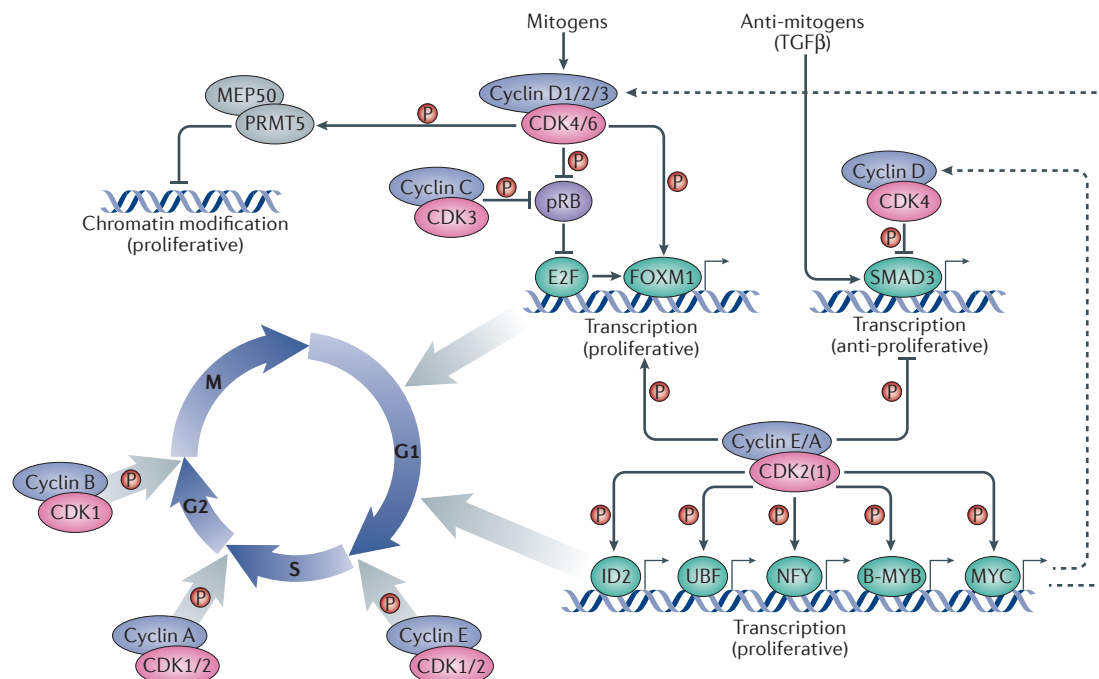


Figure 1 | Canonical roles of cyclins and cyclin-dependent kinases in the cell division cycle. Cell cycle progression is driven by heterodimeric complexes formed by cyclin D, cyclin E, cyclin A or cyclin B with cyclin-dependent kinase 4 (CDK4), CDK6, CDK2 or CDK1. Cyclin C–CDK3 complexes also participate in G0 to G1 phase progression (not shown), although their physiological relevance in this process is less well established. Cell cycle entry is driven by the formation of cyclin D–CDK4/6 complexes in response to mitogenic stimulation. These complexes phosphorylate and partially inactivate pRB, thereby releasing its inhibition of the E2F transcription factors. E2Fs subsequently promote the expression of multiple cell cycle genes, among which are those encoding E-type cyclins, which bind to and activate CDK2 (and, to a lesser extent, CDK1). Activation of CDK2 (and CDK1) by E-type and A-type cyclins leads to the phosphorylation of various transcription factors, including the helix–loop–helix (HLH) protein inhibitor of DNA binding 2 (ID2), as well as upstream binding factor (UBF), nuclear factor Y (NFY), B-MYB and MYC, which contribute to cell cycle progression at different levels. Cyclin D1–CDK4 complexes also phosphorylate chromatin modifiers such as the protein Arg N-methyltransferase 5 (PRMT5) cofactor methylome protein 50 (MEP50) to promote the repression of specific genes with anti-proliferative properties. Several cyclin–CDK complexes can phosphorylate the transcription factor SMAD3, which is activated by anti-mitogens such as transforming growth factor- β (TGF β) and forkhead box protein M1 (FOXO1), to promote cell cycle progression. Entry into mitosis is specifically driven by cyclin B–CDK1 complexes. Dashed arrows indicate indirect connections or connections with multiple steps.

which regulates the expression of various cell cycle regulators, including proteins that govern the G2–M transition¹². Cyclin D1–CDK4-mediated phosphorylation of methylome protein 50 increases the activity of the chromatin modifier PRMT5 (protein Arg N-methyltransferase 5), which is thought to mediate key events associated with cyclin D1–CDK-dependent neoplastic growth¹³.

In addition, E-type and A-type cyclins have been implicated in directly modulating transcription through CDK-dependent phosphorylation of several DNA-binding factors, including SMAD3 and FOXO1 (REFS 11, 14), inhibitor of DNA binding 2 (ID2), upstream-binding factor (UBF) and nuclear transcription factor Y (NFY)^{15–20} (FIG. 1). Cyclin E/A–CDK2 complexes also phosphorylate the transcription factors MYC and B-MYB, thereby enhancing the ability of these proteins to activate target genes during the cell cycle^{21,22}. In tumours driven by MYC overexpression, CDK2 activity seems to be essential for preventing senescence and allowing immortalization of cancer cells²³. However,

whereas the functional link between cyclin–CDKs and pocket proteins has been firmly established, the contribution of these additional phosphorylation events to normal cell cycle progression remains unknown.

Kinase-independent transcriptional roles

In addition to the role of cyclin–CDK complexes in phosphorylating transcriptional regulators, there is substantial evidence for transcriptional functions of cyclins that do not involve kinase activity. Interestingly, reports of cell cycle-independent transcriptional roles of cyclins and CDKs have, so far, been vastly dominated by D-type cyclins.

D-type cyclins have direct roles in transcription. The first indication of a pRB-independent role of cyclins in transcriptional control came from the demonstration that ectopic expression of cyclin D1 abrogated the ability of the myogenic master regulator myoblast determination protein (MYOD) to activate the muscle creatine kinase (MCK) promoter²⁴. In skeletal myoblasts,

Chromatin modifier

A protein complex that, through catalysing specific histone or DNA modifications, induces changes in chromatin to make it accessible (open) or inaccessible (closed) to the transcriptional machinery.

Table 1 | **Representative transcription factors regulated by interphase cyclins and CDKs***

Transcription factor	Cyclin	CDK	Kinase-dependent?	Activation or repression	Biological function(s)	Refs
Regulation by cyclin–CDK complexes						
B-MYB	A2	CDK2	Yes	Activation	Promoting cell cycle progression	21
MYC	E1	CDK2	Yes	Activation	Repression of cellular senescence	22
FOXM1C	D1, E1, A2	CDK2, CDK4	Yes	Activation	Promoting cell cycle progression, repression of cellular senescence	12,14
FOXO1	E	CDK2	Yes	Repression	Mediating genotoxic responses	93
FOXP3	E	CDK2	Yes	Repression	Development of regulatory T cells	128
GATA4	D1	CDK4	Yes	Repression	Preventing cardiomyocyte differentiation	103
ID2	E1, A2	CDK2	Yes	Repression	Promoting cell cycle progression	15
MEF2	D3 (D1)	CDK4	Yes	Repression	Preventing muscle differentiation	102
NFY	A2	CDK2	Yes	Activation	Promoting cell cycle progression	17,18
NGN2	A	CDK2, CDK1	Yes	Repression	Inhibition of neuronal differentiation	113
NRF1	D1	CDK4	Yes	Repression	Inhibition of mitochondrial function	141
PPAR γ	D3	CDK4	Yes	Repression	Induction of adipogenesis	109,154
RUNX2	D1	CDK4	Yes	Repression	Inhibition of bone differentiation	98
SMAD3	–	CDK2, CDK4	Yes	Repression	Preventing TGF β signalling	11
UBF	E1, A2	CDK2	Yes	Activation	Ribosomal DNA transcription	16
Regulation by cyclins						
RUNX1 (AML1)	D3	–	No	Repression	Inhibition of myeloid differentiation	46
AR	D1, D3	–	No	Repression	Inhibition of hormone-dependent gene expression	33,36
NEUROD1	D1	–	No	Repression	Inhibition of neuronal differentiation	39
CEBP β	D1	–	No	Activation	Promoting mammary epithelial differentiation	172
CBP	D1	–	No	Activation	Promoting NOTCH-dependent retinal development	36,48
DMP1	D2 (D1, D3)	–	No	Repression	Modulation of the p53 pathway	42
ER	D1	–	No	Activation	Activation of hormone-dependent gene expression	30–32
FOXO1, O3	D1	–	No	Repression	Inhibition of anoikis	97
ATF5	D3	–	No	Activation	Various cellular processes	45
HNF4 α	D1	–	No	Repression	Inhibition of lipogenesis in the liver	152
MYOD	D1	CDK4	No	Repression	Blocking myoblast differentiation	24,25,27
PPAR γ	D1	–	No	Repression	Blocking adipogenesis	38
RUNX3	D1	–	No	Repression	Cell proliferation	40
SP1	D1	–	No	Repression	Various cellular processes	28
STAT3	D1	–	No	Repression	Cytokine signalling	173
TR	D1	–	No	Repression	Hormone-dependent gene expression	41
VDR	D3	–	No	Activation	Vitamin-dependent gene expression	44
Regulation by CDKs						
JUN	–	CDK6	No	Activation	Stimulation of angiogenesis	54
RUNX1	–	CDK6	No	Repression	Inhibition of myeloid differentiation	53
STAT3	D1,2,3	CDK6	No	Activation	Cell cycle inhibition	54

AR, androgen receptor; ATF5, activating transcription factor 5; CBP, CREB-binding protein; CDK, cyclin-dependent kinase; CEBP β , CCAAT/enhancer binding protein- β ; DMP1, dentin matrix acidic phosphoprotein 1; ER, oestrogen receptor; FOX, forkhead box protein; HNF4 α , hepatocyte nuclear factor 4 α ; ID2, inhibitor of DNA binding 2; MEF2, myocyte-specific enhancer factor 2; MYOD, myogenic differentiation protein (also known as myogenic determination protein); NEUROD1, neuronal differentiation 1; NFY, nuclear transcription factor Y; NGN2, neurogenin 2; NRF1, nuclear respiratory factor 1; PPAR γ , peroxisome proliferator-activated receptor- γ ; RUNX, runt-related transcription factor; STAT3, signal transducer and activator of transcription 3; TGF β , transforming growth factor- β ; TR, thyroid hormone receptor; UBF, upstream binding factor; VDR, vitamin D receptor. *The pRB, p107 and 130 proteins are not included.

cyclin D1 recruits CDK4 to MYOD and thereby blocks MYOD-dependent transcription independently of CDK4 catalytic activity^{25–27}. Cyclin D1 was also shown to repress transcription factor SP1-mediated transcription by binding to the TATA box-binding protein (TBP)-associated factor TAFII250 in a pRB-independent manner^{28,29}. A direct interaction between cyclin D1 and the oestrogen receptor has been reported to augment oestrogen receptor-driven transcription, independently of the kinase activity of CDKs. Specifically, cyclin D1 was shown to function as a scaffold between the oestrogen receptor and members of the steroid receptor co-activator family^{30,31}, as well as between the oestrogen receptor and the chromatin modifier histone acetyltransferase PCAF (also known as KAT2B)³². By contrast, cyclin D1 and cyclin D3 inhibit the transactivation activity of the androgen receptor, which is another member of the nuclear steroid hormone receptor family, in a kinase-independent manner by competing with it for PCAF binding^{33–36}.

Numerous reports have subsequently described interactions of D-type cyclins with many transcription factors (TABLE 1). Several groups have demonstrated, mostly using protein-overexpression systems, that D-type cyclins interact with sequence-specific transcription factors and activate or repress transcription by recruiting chromatin modifiers to gene promoters (or, in some cases, by blocking their access). Cyclin D1 directly binds to and inhibits histone acetyltransferases such as p300 and CREB-binding protein (CBP) — in addition to PCAF — and represses their ability to activate various promoters, including that of peroxisome proliferator-activated receptor- γ (PPAR γ), a master regulator of adipogenesis^{32,37}. Consistent with these data, mouse embryonic fibroblasts lacking cyclin D1 have increased PPAR γ activity and are prone to adipogenic differentiation³⁸. Cyclin D1 can also repress transcription through neurogenic differentiation factor 1 (NEUROD1) or runt-related transcription factor 3 (RUNX3), independently of CDK activity, by preventing the interaction of these transcription factors with p300 (REFS 39,40). Furthermore, cyclin D1 can inhibit the transcriptional activity of thyroid hormone receptors by recruiting histone deacetylases to these transcription factors and by promoting the formation of transcriptional repressive complexes⁴¹. Cyclin D2 and cyclin D3 can also modulate the activity of several transcription factors by interacting directly with them. A yeast two-hybrid screen identified the MYB-like transcription factor DMP1 (dentin matrix acidic phosphoprotein 1) as a cyclin D2 interactor⁴². Enforced expression of any D-type cyclin inhibits the transcriptional activity of DMP1 in a CDK-independent manner^{42,43}. Intriguingly, DMP1 regulates the expression of p19^{ARF} — which is an inhibitor of E3 ubiquitin-protein ligase MDM2, a negative regulator of p53 — raising the possibility that D-type cyclins can modulate both the pRB pathway (through CDK activity) and the p53 pathway (through CDK-independent activity).

Cyclin D3 was reported to interact with and enhance the transactivation of the vitamin D receptor, and this effect was diminished by overexpression of CDK4 or

CDK6 (REF. 44). Additionally, cyclin D3 interacts with human activating transcription factor 5 (ATF5) and enhances its activity⁴⁵. However, cyclin D3 also competes with core binding factor- β for binding to RUNX1 (also known as AML1), thereby reducing the ability of RUNX1 to bind to DNA and activate transcription⁴⁶.

Collectively, these reports suggest that D-type cyclins might exert transcriptional effects independently of any associated kinase activity. To further address this, human cancer cells were engineered to overexpress wild-type cyclin D1 or a mutant cyclin D1 that is unable to activate CDK4 and CDK6 (REF. 47). Surprisingly, the set of induced target genes was identical in both cases and lacked E2F target genes, indicating that cyclin D1 has a kinase-independent and E2F-independent role in transcription. Further computational analysis implicated the transcription factor CCAAT/enhancer-binding protein β (CEBP β) as the effector of cyclin D1 in regulating gene expression. Proteomic studies using a knock-in mouse strain expressing a tagged version of cyclin D1 in place of the wild-type protein showed that cyclin D1 interacts *in vivo* with several proteins that are involved in transcription⁴⁸. Furthermore, chromatin immunoprecipitation coupled to microarray (ChIP-on-chip) analysis revealed an extensive interaction of cyclin D1 with the mouse genome⁴⁸. In particular, within developing mouse retinas, cyclin D1 physically occupies the *Notch1* gene enhancer, to which it recruits CBP acetyltransferase, thereby activating gene expression. Genetic ablation of cyclin D1 decreased recruitment of CBP to the *Notch1* gene enhancer and diminished histone acetylation and expression of NOTCH1 in the retina. Given the essential role of NOTCH1 in retinal development⁴⁹, these findings are likely to explain, at least in part, the retinal developmental abnormalities that are seen in cyclin D1-null animals^{50,51}. A direct interaction of cyclin D1 with chromatin was also demonstrated by a ChIP-sequencing study in which tagged cyclin D1 was ectopically expressed in cyclin D1-null mouse embryonic fibroblasts. The authors concluded that cyclin D1 has a role in chromosomal stability through binding to genes that govern chromosomal integrity⁵².

Direct transcriptional functions of CDK6.

Although the majority of reports have highlighted a CDK4/6-independent function of D-type cyclins in regulating transcription, some intriguing studies revealed a kinase-independent role for CDK6 itself in this process. CDK6 was shown to physically interact with and inhibit the transcriptional activity of RUNX1 in a kinase-independent fashion and, by doing so, block myeloid differentiation⁵³. CDK6, but not CDK4, activates JUN and signal transducer and activator of transcription 3 (STAT3) to induce transcription of cyclin-dependent kinase inhibitor 2a (*Cdkn2a*; encoding the CDK4/6 inhibitor p16^{INK4A}) and vascular endothelial growth factor A (*Vegfa*). Whereas the effect of CDK6 on p16^{INK4A} expression requires the presence of a D-type cyclin, the effect on VEGFA expression is independent of cyclin D⁵⁴. Interestingly, cyclin D1 has also been postulated to contribute to VEGFA gene transcription⁵⁵,

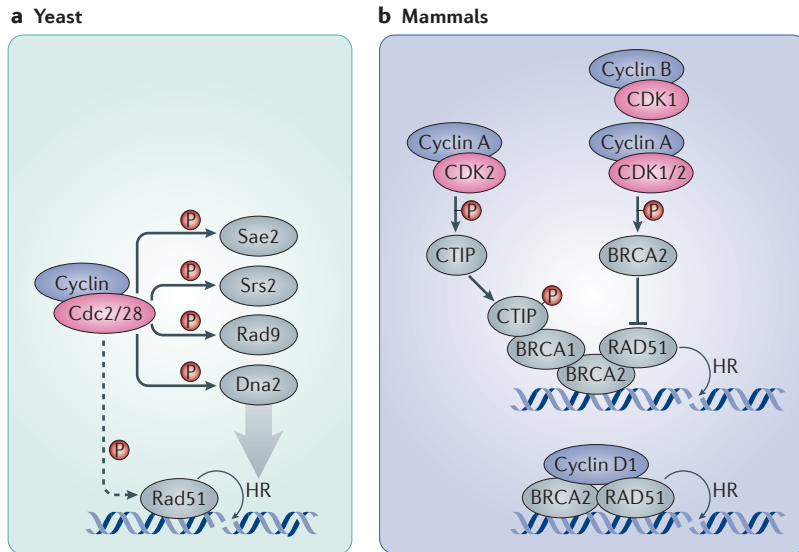


Figure 2 | Cyclins and cyclin-dependent kinases in DNA double-strand break repair. **a** | In yeast, the sole cyclin-dependent kinase (CDK) — Cdc28 in *Saccharomyces cerevisiae* and Cdc2 in *Schizosaccharomyces pombe* — participates in the recruitment of Rad51 to DNA double-strand breaks (DSBs) and phosphorylates several proteins that are involved in the DNA damage response and the repair of DSBs through homologous recombination, such as the endonuclease Sae2, the helicase Srs2, the G2 checkpoint protein Rad9 and the resection nuclease Dna2. **b** | In mammalian cells, CDK2 and CDK1 (together with A-, B- and potentially also E-type cyclins) modulate homologous recombination by the direct phosphorylation of breast cancer type 2 susceptibility protein (BRCA2), preventing it from binding to and recruiting RAD51. Conversely, CDK2 can support homologous recombination by phosphorylating CTBP-interacting protein (CTIP; the human homologue of Sae2), which promotes the recruitment of BRCA1 to DSBs. In addition, through binding to BRCA2 at DSBs, cyclin D1 helps to recruit RAD51 to DSB sites by directly interacting with it in a CDK-independent manner, thereby promoting homologous recombination. Dashed arrows indicate indirect connections or connections with multiple steps. HR, homologous recombination.

suggesting that both CDK6 and cyclin D1 may regulate angiogenesis through different routes. The observation that CDK6 upregulates its own inhibitor p16^{INK4A} indicates the presence of a negative feedback loop, by which CDK6 safeguards against uncontrolled proliferation triggered by CDK6 overexpression⁵⁴. CDK6 can exert its pro-proliferative role only upon silencing of the gene encoding p16^{INK4A}, an event that is frequently seen in human tumours⁶. Additional data suggest that the transcriptional activity of CDK6 may also be crucial in regulating the balance between quiescence and proliferation in haematopoietic and leukaemic stem cells by modulating the activity of the transcription factor early growth response protein 1 (EGR1)⁵⁶, although whether CDK6 binds directly to EGR1 is currently unknown.

Cyclins and CDKs in DNA damage repair

DNA double-strand breaks (DSBs) are repaired by two distinct mechanisms: the high-fidelity homologous recombination, which uses the sister chromatid as the main template for repair; and non-homologous end-joining (NHEJ), which directly joins the broken DNA and is therefore error-prone. Work in several systems indicates that components of the core cell cycle machinery have important and direct roles in DSB repair.

Yeast. In *Saccharomyces cerevisiae*, inhibition of the sole cyclin-dependent kinase Cdc28 (which is homologous to CDK1) after the induction of DSBs causes defects in end-resection of damaged DNA, thereby preventing the loading of the homologous recombination repair proteins Rad51 and replication protein A to sites of DNA damage^{57,58}, when Cdc28 is inhibited before the induction of DSBs, the incidence of error-prone NHEJ increases substantially, indicating that the kinase activity of Cdc28 is necessary to support homologous recombination⁵⁷ (FIG. 2a). In *Schizosaccharomyces pombe*, the CDK1 homologue Cdc2 promotes the formation of Rad51 foci, resulting in the formation of recombination intermediates at early stages of homologous recombination. At a later stage of repair, Cdc2 and a checkpoint protein Crb2 were shown to control topoisomerase III activity, thereby ensuring complete processing of recombination intermediates⁵⁹. *S. cerevisiae* Cdc28 can also regulate DNA damage repair by phosphorylating various substrates, such as Sae2, an endonuclease that is required for the initiation of resection of single-stranded DNA⁶⁰. Another substrate of the *S. cerevisiae* Cdc28 is the Srs2 helicase, the phosphorylation of which prevents its sumoylation and targets it to dismantle specific DNA structures, such as D-loops, in a helicase-dependent manner during homologous recombination⁶¹. During DNA damage repair, Cdc28 also phosphorylates the G2 checkpoint protein Rad9 (REFS 62–64) and the resection nuclease Dna2 (REF. 65). Collectively, these studies present a strong case for the requirement of CDK activity in the resection of DSB ends and in mediating the choice between the homologous recombination and NHEJ repair pathways in yeast⁶⁶.

Mammals. In contrast to yeast, cyclin A- and cyclin B-associated CDK activity was found to inhibit homologous recombination in mammalian cells. Cyclin A-CDK1/2 and cyclin B-CDK1 complexes phosphorylate Ser3291 of breast cancer type 2 susceptibility protein (BRCA2)⁶⁷, thereby preventing it from binding to and recruiting RAD51 to sites of DNA damage (FIG. 2b). DNA damage leads to reduced levels of Ser3291 phosphorylation and restoration of the interaction between BRCA2 and RAD51 (REF. 67). Phosphorylation of Ser3291 is low in S phase but increases towards mitosis, suggesting that the termination of homologous recombination by CDKs is necessary for normal chromosome segregation. Consistent with this, expression of mutant forms of BRCA2 that cannot be phosphorylated by CDK does not change the rate of homologous recombination but delays the onset of mitosis⁶⁸. CDK2 was also shown to support homologous recombination by phosphorylating C-terminal binding protein (CTBP)-interacting protein (CTIP; the human homologue of Sae2), thereby promoting its interaction with BRCA1 and the MRE11 exonuclease and leading to the resection of DSBs⁶⁹ (FIG. 2b).

As a consequence of DNA damage, cyclin D1 is degraded and cyclin D-CDK4/6 complexes are disrupted, which contributes to cell cycle arrest⁷⁰. Expression of degradation-resistant mutants of cyclin D1 results in

Quiescence

A post-mitotic state in which cells do not divide, although they may be induced to re-enter the cell cycle in response to specific stimuli.

D-loops

Displacement loops formed in homologous recombination repair. Two strands of a homologous chromosome are separated for a stretch by an invading broken strand, forming a displaced single-stranded structure, the D-loop.

Apoptosis

A form of programmed cell death characterized by the activation of specific proteases, termed caspases, which induce chromatin fragmentation and the degradation of various cellular components.

increased levels of DNA damage, confirming the importance of preventing cell cycle progression in these conditions⁷¹. An unexpected direct role for cyclin D1 in DNA damage repair was first suggested by the observation that targeting cyclin D1 to chromatin led to co-recruitment of RAD51 (REF. 72). Subsequently, it was reported⁷³ that cyclin D1 localizes — in a BRCA2-dependent manner — to irradiation-induced DSBs in human cancer cells, where it helps to recruit RAD51, thereby stimulating homologous recombination (FIG. 2b). Depletion of cyclin D1 in human cancer cells reduced the recruitment of RAD51, decreased homologous recombination and rendered cells more sensitive to DNA damage. Notably, this function of cyclin D1 was shown to be kinase-independent and to also take place in pRB-deficient cells⁷³.

A positive role for cyclin D1 in DNA damage repair might seem to be inconsistent with the observation that cyclin D1 is rapidly degraded after DNA damage. However, a relatively small pool of cyclin D1 persists after irradiation, and cyclin D1 becomes redistributed to DNA damage sites. This DNA-bound pool of cyclin D1 is likely to have a positive role in DNA damage repair⁷³. On the other hand, forced overexpression of cyclin D1 might override cell cycle arrest following DNA damage, leading to DNA replication stress and, ultimately, apoptosis. Consistent with this notion, overexpression

of cyclin D1 in cultured tumour cells rendered the cells more sensitive to radiation and triggered apoptosis^{74,75}. One important and unresolved issue is how overexpression of cyclin D1, which is frequently observed in human cancers, affects the response of tumour cells to DNA damage *in vivo*.

Balancing cell proliferation and death

The regulation of the cell cycle is tightly linked to the control of cell death. Indeed, early studies indicated that the pRB–E2F pathway could modulate the expression of multiple pro- or anti-apoptotic proteins^{5,76}. Overexpression of cyclins in cultured cells leads to either increased apoptosis or increased survival, depending on the cellular setting. For instance, ectopic expression of cyclin D1 induces cell death and increases sensitivity to cytotoxic agents in various cell lines^{74,77–81}. By contrast, overexpression of cyclin D1 protects fibrosarcoma cells from cell death⁸², and suppression of cyclin D1 using antisense oligonucleotides sensitizes pancreatic cancer cells to apoptosis induced by cytotoxic agents⁸³. Similarly, ectopic expression of cyclin D3 in leukaemic T cells prevents them from apoptosis that would normally be induced by phorbol myristate acetate⁸⁴, and ablation of cyclin D3 in T cell acute leukaemia cells triggers cell death⁸⁵. Whether these effects are direct or mediated by the pRB–E2F pathway is unclear in most of the cases. However, D-type cyclins and CDKs can also modulate apoptotic cell death through direct interaction with the apoptotic machinery. In mantle cell lymphoma, cyclin D1, which is frequently overexpressed in this tumour type, binds to and sequesters the pro-apoptotic protein BAX in the cytoplasm, thereby inhibiting apoptosis⁸⁶ (FIG. 3). CDK4 interacts with the apoptosis inhibitor survivin (also known as baculoviral IAP repeat-containing protein 5). As a result, p21^{CIP1} is released from its complex with CDK4 and interacts with mitochondrial pro-caspase 3 to suppress FAS-mediated cell death⁸⁷.

A pro-apoptotic function for cyclin D3 is suggested by the observation that it can physically interact with caspase 2, leading to the stabilization of this protease and promoting its cleavage into an active form⁸⁸ (FIG. 3). Cyclin D1–CDK kinase activity might play a part in mediating cell death under pathological conditions. Increased activity of cyclin D1–CDK4, correlating with neuronal death, has been reported in a rat model of stroke⁸⁹. Moreover, neuronal cell death triggered by treatment of rats with kainic acid can be prevented by injection of CDK4 or cyclin D1 antisense oligonucleotides into brain ventricles⁹⁰. Administration of flavopiridol, which is a pharmacological inhibitor of CDKs, into brain ventricles prevents neuronal cell death in a model of focal ischaemia⁸⁹.

Analyses of mice lacking D-type cyclins support the pro-survival function of these proteins in certain tissues. Genetic ablation of cyclin D1 leads to increased apoptosis of photoreceptor cells in postnatal animals⁹¹. Whereas constitutive, germline ablation of cyclin D1 (or cyclin D2 or D3) does not cause obvious cell death in other organs, an acute simultaneous shutdown of all three D-type cyclins in the bone marrow of adult mice

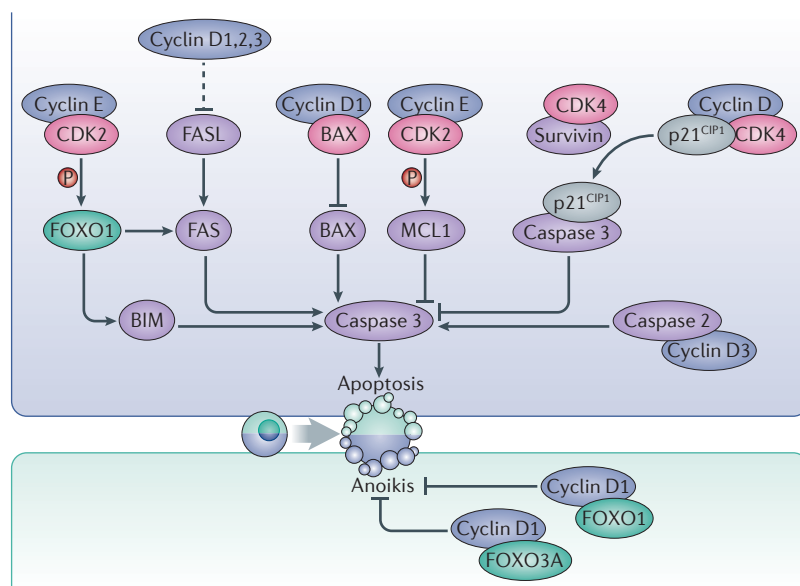


Figure 3 | Regulation of cell death by interphase cyclins and cyclin-dependent kinases. Interphase cyclins modulate apoptosis using cyclin-dependent kinase (CDK)-dependent and CDK-independent mechanisms. Cyclin E–CDK2 complexes phosphorylate the pro-survival factor induced myeloid leukaemia cell differentiation protein MCL1, as well as the transcription factor forkhead box protein O1 (FOXO1), leading to the inhibition or activation of apoptosis, respectively. Phosphorylation of MCL1 stabilizes the protein, whereas phosphorylation of FOXO1 leads to upregulated expression of the pro-apoptotic factors FAS and BIM. Binding of survivin to CDK4 causes the release of the CDK inhibitor p21 (CIP1), which binds to and inhibits caspase 3. D-type cyclins can also directly interact with the pro-apoptotic proteins BAX or caspase 2. Moreover, the three D-type cyclins have overlapping roles in repressing the expression of the death receptor FAS and its ligand FASL, thereby preventing apoptosis in haematopoietic cells. Cyclin D1 also interacts with FOXO1 and FOXO3A and prevents anoikis. Dashed arrows indicate indirect connections or connections with multiple steps.

Box 1 | Cyclin-dependent kinase inhibitors

Two different families of proteins can inhibit cyclin-dependent kinase (CDK) activity¹⁵⁹. Members of the CIP/KIP (CDK-interacting protein/kinase inhibitory protein) family of inhibitors — p21 (CIP1), p27 (KIP1) and p57 (KIP2), encoded by the cyclin-dependent kinase inhibitor 1 (CDKN1) genes — bind to most cyclin-CDK complexes and prevent their activation. The CDKN2-encoded p16 (INK4A), p15 (INK4B), p18 (INK4C) and p19 (INK4D) proteins, which constitute the INK4 family, bind to CDK4 and CDK6 and block their association with the D-type cyclins. Owing to their inhibition of CDKs and cell proliferation, CDK inhibitors are tumour suppressors, and deletion of the CDKN2A-CDKN2B locus, which encodes p16, p15 and another protein p14 (ARF), which is involved in the activation of the p53 tumour suppressor, is one of the most frequent mutations in human tumours¹⁶⁰.

CDK inhibitors are involved in many cellular processes, including cell cycle control, transcription, cell migration and metabolism, and they can affect cell quiescence, senescence, differentiation or cell death, depending on the cellular setting^{112,143,155,161–164}. In most cases, these functions are mediated by inhibition of specific cyclin-CDK complexes. However, INK4 or CIP/KIP proteins can have additional activities that are unrelated to their function as CDK inhibitors¹⁶⁵. For example, p19^{INK4D} has been proposed to participate in DNA damage repair and senescence in a CDK-independent manner through directly interacting with chromatin^{166,167}. p21^{CIP1} can function as a RHO-kinase (ROCK) inhibitor, and p57^{KIP2} modulates the subcellular localization of LIM kinase (LIMK), a Ser/Thr kinase involved in the regulation of actin filaments¹⁶⁸. Similarly, p27^{KIP1} has an important role in the control of cytoskeleton assembly and migration¹⁶⁹ by physically interacting with RHOA and by blocking its activation, resulting in impaired stabilization of actin fibres and increased cell motility^{170,171}.

triggers massive apoptosis. In this compartment, D-type cyclins repress the expression of death receptor FAS and its ligand FASL (FIG. 3); ablation of the three D-type cyclins results in upregulation of FAS and FASL proteins, leading to haematopoietic cell apoptosis⁹².

Evidence for a direct role of cyclin E-CDK2 or cyclin A-CDK2 complexes in the control of cell death is more limited. Cyclin E-CDK2 affects the apoptotic response through phosphorylation of the transcription factor FOXO1, thus upregulating the expression of the pro-apoptotic proteins FAS and BIM (also known as BCL2L11)⁹³. However, it has also been suggested that cyclin E-CDK2 complexes promote survival by directly phosphorylating the anti-apoptotic factor induced myeloid leukaemia cell differentiation protein MCL1 (REF. 94) (FIG. 3).

Cyclins and CDKs can modulate other cell death pathways in addition to apoptosis. Cyclin D1 expression suppresses autophagy in the mammary epithelium⁹⁵, a function that may be mediated by the pRB-E2F pathway⁹⁶. In prostate cancer cell lines, cyclin D1 was postulated to inhibit anoikis by binding to FOXO1 and FOXO3A and blocking their ability to induce anoikis in a CDK-independent manner⁹⁷ (FIG. 3).

Control of cell differentiation

Terminal differentiation is usually coupled to permanent exit from the cell cycle. The levels of cyclins typically decline when cells exit the cell cycle and undergo differentiation. Moreover, induction of the expression of CDK inhibitors during cell differentiation prevents activation of cyclin-CDK complexes in terminally differentiated cells (BOX 1). On the other hand, expression of cyclin-CDK complexes in proliferating cells inhibits pRB function, thereby promoting proliferation and

inhibiting differentiation. Indeed, pRB — in addition to its well-established role in the cell cycle — binds to and regulates the activity of several cell type-specific transcription factors, including MYOD, myocyte-specific enhancer factor 2 (MEF2), RUNX2 and many others, thereby linking cell cycle arrest and differentiation⁵.

Cyclins and CDKs have also been shown to carry out direct, pRB-independent functions in cell differentiation. For example, cyclin D1-CDK4 complexes phosphorylate RUNX2, a transcription factor that drives osteoblast differentiation, and target it for degradation, thereby inhibiting bone differentiation⁹⁸. It is likely that CDK1 and CDK2 can affect cell differentiation through phosphorylation and activation of EZH2, the catalytic subunit of the histone Lys methyltransferase Polycomb repressive complex 2 (PRC2). PRC2 is an important regulator of gene expression through catalysing the trimethylation of histone H3 Lys 27 (H3K27me3), and it influences differentiation by repressing lineage-specific genes^{99,100}.

Roles of cyclins and CDKs in muscle differentiation.

As mentioned above, cyclin D1 and CDK4 can prevent transcriptional activation mediated by MYOD and hence inhibit myoblast differentiation^{24–27,101} (FIG. 4a). Although this activity is partially kinase-independent, maximal repression of MYOD, as well as repression of another muscle-specific basic helix-loop-helix (bHLH) transcription factor, myogenin, requires CDK kinase activity¹⁰². Cyclin D-CDK4 complexes also block the association of MEF2 with glutamate receptor-interacting protein 1 (GRIP1), which is a member of the steroid receptor family of transcription co-activators, thereby inhibiting the ability of MEF2 to induce muscle gene expression¹⁰². This block requires CDK activity, although the exact molecular mechanism by which it is mediated is unclear¹⁰². Cyclin D1 can also inhibit the differentiation of cardiomyocytes through CDK4-dependent phosphorylation of GATA4, which targets this transcription factor for degradation¹⁰³ (FIG. 4a).

By contrast, specific D-type cyclins are upregulated during terminal differentiation of certain cell types. In differentiating skeletal myoblasts, the expression of cyclin D3 is strongly upregulated (over 20-fold), in part by MYOD^{101,104,105}. In differentiated myotubes, cyclin D3 becomes stabilized by binding to pRB¹⁰⁶ (FIG. 4a). High levels of cyclin D3 have also been observed in several other terminally differentiated cell types, including many types of epithelia, oocytes and ovarian corpora lutea, podocytes within renal glomeruli, adipocytes (see below) and cells of the exocrine pancreas^{107–110}. On the basis of these observations, cyclin D3 has been postulated to carry out cell cycle-independent functions in a range of terminally differentiated cells. However, these findings seem to be in conflict with the observation that mice lacking cyclin D3 develop normal muscle and epithelia¹¹¹. This might be attributed to functional redundancy between the three D-type cyclins, and conditional knockout approaches involving deletion of all D-type cyclins in post-mitotic tissues may help to clarify this point.

Autophagy

The degradation and recycling of cellular components and organelles by lysosomes.

Anoikis

A special form of programmed cell death that can be induced in anchorage-dependent cells by their detachment from the surrounding matrix.

Basic helix-loop-helix (bHLH) transcription factor

A protein that contains two helices separated by a loop (the HLH domain), which binds to DNA in a sequence-specific manner.

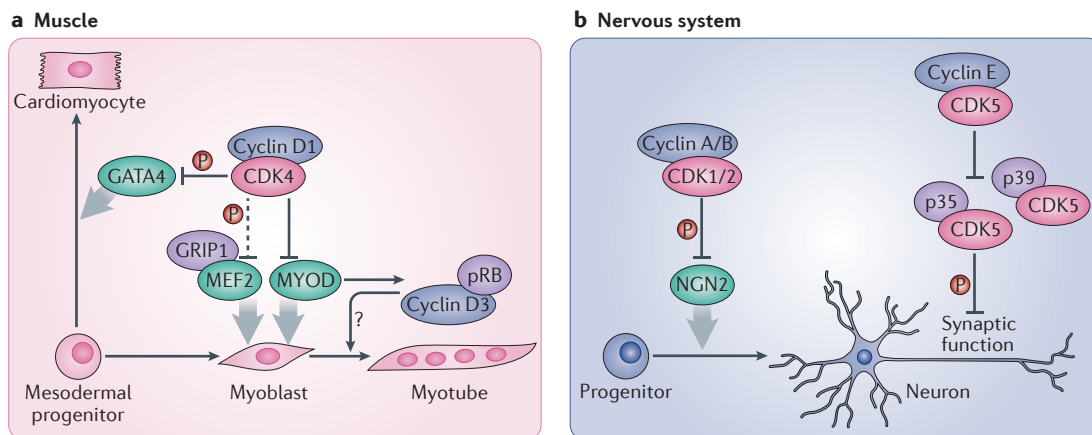


Figure 4 | Roles of interphase cyclins and cyclin-dependent kinases in cell differentiation. a | In muscle progenitor cells, cyclin D1 interacts with the transcription factor myoblast determination protein (MYOD) and inhibits its activity. This effect requires cyclin-dependent kinase 4 (CDK4) but is partially independent of its kinase function. However, the catalytic activity of CDK4 is required for the inhibition by cyclin D1–CDK4 of GRIP1–MEF2 (glutamate receptor-interacting protein 1–myocyte-specific enhancer factor 2) complexes, which are also involved in muscle differentiation. Cyclin D1–CDK4 complexes also phosphorylate and inactivate the GATA4 transcription factor, and thereby repress the differentiation of cardiomyocytes. Conversely, cyclin D3, which is stabilized in differentiated muscle cells by its interaction with pRB, seems to promote muscle cell differentiation by an unknown mechanism. **b** | In the nervous system, the differentiation of neuronal progenitors into neurons is blocked by cyclin A–CDK2 complexes, which phosphorylate and inactivate the neurogenin 2 (NGN2) transcription factor. Cyclin E regulates synaptic plasticity by forming kinase-inactive complexes with CDK5 and sequestering it from its activators p35 and p39, thereby inhibiting the phosphorylation of synaptic CDK5 substrates. Dashed arrows indicate indirect connections or connections with multiple steps.

Cyclin D–CDKs in neuronal development. Cyclin–CDK complexes can inhibit neuronal differentiation, not only through the pRB–E2F pathway¹¹² but also by phosphorylating and inactivating transcription factors that drive neurogenesis. For instance, neurogenin 2 (NGN2), which is a master regulator of neuronal differentiation, is inhibited by multi-site phosphorylation by CDKs^{113,114} (FIG. 4b), and a non-phosphorylatable NGN2 mutant can drive neuronal differentiation in the presence of CDK activity¹¹³. Cyclin D1–CDK4 activity in neural stem cells is also thought to inhibit neurogenesis and expand the population of basal progenitors by shortening the duration of G1, although the exact mechanism remains to be clarified¹¹⁵. In support of a role for cyclin–CDK complexes in inhibiting neuronal differentiation, embryos from *Cdk2*^{−/−}*Cdk4*^{−/−} knockout mice lack the intermediate zone and cortical plate in their brains, owing to premature neuronal differentiation and depletion of basal progenitor cells¹¹⁶.

Other data support the opposite conclusion — namely, that D-type cyclins promote neurogenesis. A cell cycle-independent function for cyclins in the nervous system was first proposed based on the upregulation of cyclin D1-associated kinase activity in rat brains at the onset of neuronal differentiation^{117,118}. High levels of cyclin D1 persisted in the brains of adult mice, whereas the levels of cyclin D2 decreased, suggesting a specific role for cyclin D1 in neuronal differentiation. By contrast, another study observed high levels of cyclin D2 in the brains of adult mice, suggesting that cyclin D2 has a distinct role in terminally differentiated neurons¹¹⁹.

Loss-of-function and gain-of-function experiments in embryonic chick spinal cord also supported a role for cyclin D1 in promoting neuronal differentiation¹²⁰. *In vivo* knockdown of cyclin D1, but not cyclin D2, significantly reduced differentiation of motor neuron progenitors. Conversely, forced overexpression of cyclin D1 enhanced neurogenesis, whereas cyclins D2 and D3 inhibited this process. Interestingly, overexpression of a cyclin D1 mutant that is unable to activate CDKs promoted neurogenesis to an even greater extent than wild-type cyclin D1 (REF. 120). Collectively, these findings suggest a differential role for cyclins D1 and D2 in neurogenesis, but the molecular mechanisms involved remain unclear.

Roles of cyclin E in neuronal development. Cell cycle proteins also affect the balance between progenitors and neurons by controlling the regulation of asymmetric cell divisions of neural progenitors¹²¹. In *Drosophila melanogaster*, cyclin E is expressed in the thoracic cell lineage, which undergoes asymmetric divisions that give rise to neuroblasts (central nervous system stem cells in *D. melanogaster*) and to glial progenitors. Depletion of cyclin E renders the thoracic lineage capable of generating only glial progenitors, whereas ectopic expression of cyclin E in the abdominal lineage forces it into asymmetric divisions¹²². Mechanistically, cyclin E interacts with and inhibits the activity of Prospero, which is a transcription factor required for glial cell identity¹²³. The role of E-type cyclins in asymmetric cell divisions of mammalian neural progenitors has not been explored.

Asymmetric cell divisions
Special form of cell division in which the daughter cells have different fates, probably as a consequence of asymmetric distribution of cellular components during division.

Interestingly, cyclin E is upregulated in the brains of postnatal animals as neurons undergo terminal differentiation. In terminally differentiated neurons, cyclin E forms a complex with CDK5 (REF. 124) (FIG. 4b), which is a well-established regulator of neuronal differentiation¹²⁵. Cyclin E sequesters CDK5 into kinase-inactive cyclin E–CDK5 complexes, away from the CDK5-activating partners p35 and p39. Through this mechanism, cyclin E controls phosphorylation of key synaptic proteins and regulates synaptic plasticity and memory formation. Ablation of cyclin E in mouse brains decreased synaptic function and rendered animals unable to normally retain memories¹²⁴. These findings raise the possibility that abnormalities in cyclin E levels may underlie some cases of human learning disabilities¹²⁴.

Immune system differentiation and function. A role for CDK2 in suppressing immune tolerance was highlighted by observations that when cardiac allografts were transplanted into CDK2-null recipient mice, a long-term allograft acceptance was observed, in contrast to wild-type recipients, which rejected the grafts¹²⁶. CDK2-deficient regulatory T cells (Tregs) displayed increased suppressive capacity, probably explaining the increased immune tolerance in CDK2-null mice^{126,127}. Cyclin E–CDK2 was shown to phosphorylate and negatively regulate FOXP3, a transcription factor that is required for the development and function of Tregs¹²⁸. Furthermore, in developing lymphocytes, cyclin A–CDK2 triggers the periodic degradation of the recombinase RAG2 (V(D)J recombination-activating protein 2), thereby restricting V(D)J recombination to G1 phase¹²⁹. Mutation of CDK2-substrate residues of RAG2 abrogated cell cycle-dependent degradation of this protein¹³⁰.

Regulation of cell migration

During keratinocyte differentiation, cyclin D1 co-localizes with SEC6 (also known as EXOC3; a subunit of the exocyst complex involved in vesicle trafficking), as well as with β 1 integrin and RALA (a GTPase that regulates the exocyst). Ectopic overexpression of cyclin D1 increases recycling of β 1 integrin upon differentiation, as demonstrated by a pronounced loss of membrane-bound β 1 integrin, and leads to reduced keratinocyte adhesion to the extracellular matrix¹³¹. Cell migration and the actin cytoskeleton are also modulated by cyclins and CDKs at different levels¹³². CDK6 was shown to localize to the ruffling edges of spreading fibroblasts prior to the formation of filamentous actin and to promote migration in a $\alpha_v\beta_3$ integrin-dependent manner¹³³. Cyclin D1 was implicated in promoting fibroblast migration by inhibiting the thrombospondin 1 and RHO kinase (ROCK) signalling pathways¹³⁴. Cyclin D1 can also enhance cell migration by negatively regulating the transcription of the E3 ubiquitin ligase factor SKP2 (S-phase kinase-associated protein 2)¹³⁵. SKP2 promotes the degradation of the CDK inhibitor p27^{KIP1}, which has been implicated in cell migration through control of the RHOA pathway¹³⁶ (BOX 1). Cyclin D1 can also affect cell migration by binding to and promoting the phosphorylation of filamin A, a member of the actin-binding filamin protein

family. Knockdown of cyclin D1 or ectopic expression of p16^{INK4A} reduced the phosphorylation of filamin A and of many other actin-binding proteins, suggesting a direct role for a cyclin D1-associated kinase in regulating the cytoskeleton¹³⁷. Furthermore, it was postulated that cyclin D1 effects cell migration in a CDK-independent manner by transcriptionally regulating DICER¹³⁸, which is an endoribonuclease required for microRNA processing. In summary, although cyclin D1 has been shown to promote cell migration in several cell types, the main mechanism responsible for this function remains unclear. It also remains to be seen whether overexpression of cyclin D1, as observed in many cancer types, contributes to the increased invasiveness of tumour cells.

Roles of cyclins and CDKs in metabolism

Multiple cellular functions of cyclins and CDKs converge to control the generation of cellular energy and metabolism at both the cellular and organismal levels.

Cellular energy production and metabolism. In *D. melanogaster* and in mammalian cells, cyclin D–CDK4 activity positively regulates mitochondrial biogenesis, through pRB–E2F-dependent control of mitochondrial transcription factors such as nuclear respiratory factor 1 (NRF1) and NRF2 (REF. 139) (FIG. 5). By contrast, cyclin D1–CDK4 can also inhibit mitochondrial function: depletion of cyclin D1 in mouse breast cancer cells *in vivo* promoted expression of genes involved in mitochondrial function and increased the number of mitochondria¹⁴⁰, and cyclin D1-null mouse mammary epithelial cells had increased mitochondrial size¹⁴⁰. The inhibitory effects of cyclin D1 on mitochondrial function were attributed to mediating the repression of NRF1 through CDK-dependent phosphorylation, leading to impaired expression of mitochondrial transcription factor A (mtTFA), a factor that is essential for mitochondrial respiration¹⁴¹ (FIG. 5). Cyclin D1 was also shown to affect mitochondrial function by inhibiting the expression and activity of the glycolytic enzyme hexokinase 2 (REF. 140).

Cyclin D1 can also exert its effect on mitochondria independently of CDK4 by binding to the voltage-dependent anion channel protein (VDAC) in the outer mitochondrial membrane, thereby preventing VDAC from binding to hexokinase 2 and impairing the access of ADP to the inner mitochondrial membrane¹⁴². Further observations that cyclin D1 can directly bind to additional lipogenic enzymes and mitochondrial proteins⁴⁸ point to cyclin D1 as an important regulator of metabolic activities through different pathways¹⁴³. In addition to cyclin D1, cyclin E has been proposed to inhibit mitochondrial biogenesis and oxidative metabolism in a transcriptional manner, but the exact mechanism is unknown¹⁴⁴. Finally, the cyclin B1–CDK1 complex was shown to localize to the matrix of mitochondria, where it phosphorylates various mitochondrial proteins, including the respiratory chain complex 1 subunits, leading to increased mitochondrial respiration. It was postulated that this event coordinates mitochondrial respiration with G2–M phase progression¹⁴⁵.

Allografts

Transplants from genetically non-identical individuals of the same species.

Exocyst complex

A highly conserved, octameric protein complex that regulates vesicle delivery to, and docking at, the cell surface.

Regulation of metabolism at the organismal level.

A physiological role for cell cycle proteins in metabolism was first documented by the observations that mice lacking CDK4 developed diabetes owing to the loss of pancreatic islet β -cells^{146,147}. Conversely, knock-in mice expressing a constitutively active CDK4 mutant

developed pancreatic hyperplasia¹⁴⁶. Mice lacking cyclin D2 also had substantially reduced β -cell mass and reduced islet size, leading to glucose intolerance¹⁴⁸. This phenotype was further exacerbated by the additional heterozygous deletion of cyclin D1, and *cyclin D1*^{+/-}*D2*^{-/-} mice developed life-threatening diabetes by 3 months of age¹⁴⁹. Ablation of cyclin D3 exacerbated diabetes in non-obese diabetic mice but did not affect the proliferation rate of β -cells, raising the possibility that cyclin D3 might have a cell cycle-independent role in this tissue¹¹⁰. Glucose challenge was proposed to promote cyclin D2–CDK4 complex formation and kinase activity, which indirectly induces the expression of KIR6.2, a component of the potassium ATP channel that positively regulates insulin secretion¹⁵⁰ (FIG. 5). Conversely, inhibition of CDK4 activity leads to lower expression of KIR6.2, decreases insulin secretion and results in glucose intolerance¹⁵⁰. The effect of cyclin D2–CDK4 activity on KIR6.2 levels is mediated by E2F1, but it occurs independently of cell cycle progression¹⁵⁰. An independent mechanism linking insulin signalling to cyclin D1–CDK4 was provided by the observation that insulin-mediated upregulation of cyclin D1 (and the subsequent activation of cyclin D1–CDK4) in hepatocytes leads to phosphorylation and activation of the histone acetyltransferase GCN5 (also known as KAT2A). GCN5 is responsible for acetylation and inhibition of the transcription factor PPAR γ co-activator 1 α (PGC1 α)¹⁵¹, which inhibits the expression of gluconeogenesis genes, thereby suppressing hepatic glucose production¹⁵¹ (FIG. 5). Cyclin D1 was shown to inhibit lipid metabolism in the liver in a kinase-independent manner, through its interaction with and suppression of the transcription factor hepatocyte nuclear factor 4 α ¹⁵².

The functions of cyclins and CDKs in modulating metabolic pathways contribute to the ability of these proteins to affect differentiation. A positive role for cyclin D3 in promoting adipocyte differentiation is supported by the observation that knockdown of cyclin D3 inhibits adipogenesis *in vitro*¹⁵³. Consistent with these data, mice lacking cyclin D3 or CDK4 are protected from diet-induced obesity, and have smaller adipocytes and reduced expression of adipogenesis genes^{109,154}. CDK4 — probably in complex with cyclin D3 — also participates in adipocyte differentiation by directly interacting with and activating PPAR γ , the master regulator of adipogenesis^{109,154} (FIG. 5). Furthermore, cyclin D–CDK4 complexes may engage the pRB–E2F pathway to inhibit the pRB-mediated repression of PPAR γ ¹⁵⁵. Cyclin D1, conversely, was shown to inhibit adipogenesis by inhibiting p300 and recruiting histone deacetylases to PPAR response elements in the promoter of the lipoprotein lipase. Indeed, cyclin D1-null fibroblasts are prone to adipogenic differentiation^{37,38}.

Conclusions and perspectives

Research carried out during the past two decades indicates that interphase cyclins and CDKs have functions that extend well beyond the regulation of cell cycle progression. However, further work is needed to reconcile often-conflicting models and observations,

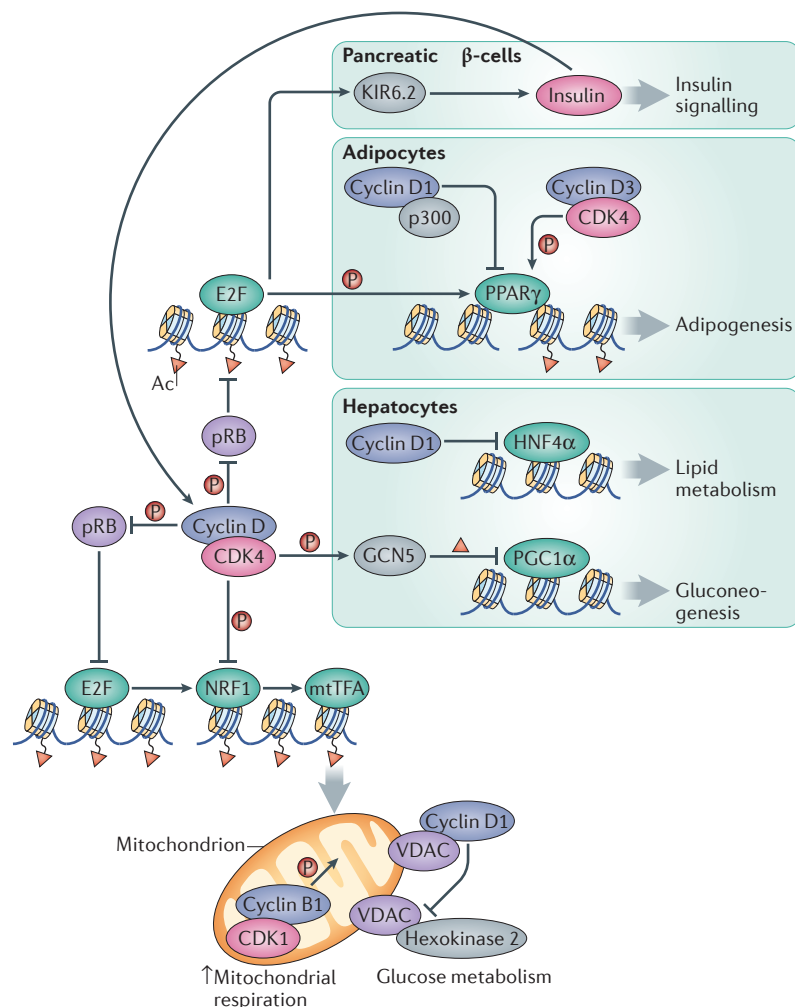


Figure 5 | An integrated view of the roles of interphase cyclins and cyclin-dependent kinases (CDKs) in the control of cellular energy and physiological metabolism.

Cyclin–CDK complexes have opposing roles in controlling mitochondrial function. Although their activity promotes the expression of nuclear respiratory factor 1 (NRF1) through the pRB–E2F pathway, cyclin D1–CDK4 can also directly phosphorylate and inactivate NRF1, leading to decreased transcription of mitochondrial proteins by mitochondrial transcription factor A (mtTFA). Moreover, the activity of mitochondria is directly stimulated by cyclin B1–CDK1, whereas binding of cyclin D1 to voltage-dependent anion channel protein (VDAC) prevents hexokinase 2 activation and the use of glucose in mitochondria. In pancreatic β -cells, cyclin D–CDK4 complexes promote insulin secretion through the E2F-mediated expression of the potassium ATP channel component KIR6.2. Insulin, in turn, induces the expression of D-type cyclins through a positive feedback loop. In adipocytes, adipogenesis is inhibited by the interaction of cyclin D1 with the histone acetyltransferase p300, which leads to inhibition of the transcription factor peroxisome proliferator-activated receptor- γ (PPAR γ). Conversely, adipogenesis is activated by cyclin D3–CDK4-mediated PPAR γ phosphorylation. Gluconeogenesis is inhibited in hepatocytes by cyclin D1–CDK4-dependent phosphorylation and activation of the acetyltransferase GCN5, which acetylates and inactivates the transcription factor PPAR γ co-activator 1 α (PGC1 α). In addition, cyclin D1 can bind to and inactivate the transcription factor hepatocyte nuclear factor 4 α (HNF4 α), resulting in deficient lipid metabolism.

a task that is perhaps confounded by the overlapping and/or compensatory roles of the different cyclin and CDK proteins in mammals. Dysregulation of cell cycle proteins is commonly found in several pathological conditions, including cancer, neurodegeneration and cardiac disease, but the relevance of non-canonical functions of cell cycle proteins in controlling processes other than cell proliferation — such as transcription, cell death, differentiation and metabolism — remains to be fully explored.

Some of the proposed non-canonical roles may not reflect a normal physiological protein function, but rather a gain-of-function event that occurs in tumour cells as a consequence of cyclin and/or CDK

overexpression. Given that several inhibitors of CDK4/6 kinase activity are currently in clinical trials^{156,157}, it will be important to elucidate which of these cell cycle-independent roles of cyclins and CDKs truly contribute to tumorigenesis *in vivo*. According to the prevailing model, the main role of CDK4 and CDK6 is to inactivate pRB. Indeed, inhibition of CDK4 and CDK6 has no effect on the proliferation of tumour cells that have lost pRB expression¹⁵⁸. Given the increasing number of reported CDK4/6 substrates, it remains to be seen whether CDK4/6 inhibition in pRB-negative tumour cells affects other non-canonical functions of cyclin D-CDK4/6 kinases and, by doing so, contributes to the therapeutic effect.

- Morgan, D. O. *The Cell Cycle: Principles of Control* (New Science Press, 2007).
- Malumbres, M. & Barbacid, M. Cell cycle, CDKs and cancer: a changing paradigm. *Nat. Rev. Cancer* **9**, 153–166 (2009).
- Malumbres, M. Cyclin-dependent kinases. *Genome Biol.* **15**, 122 (2014).
- Choi, Y. J. & Anders, L. Signaling through cyclin D-dependent kinases. *Oncogene* **33**, 1890–1903 (2014).
- Dick, F. A. & Rubin, S. M. Molecular mechanisms underlying RB protein function. *Nat. Rev. Mol. Cell Biol.* **14**, 297–306 (2013).
- Beroukhi, R. *et al.* The landscape of somatic copy-number alteration across human cancers. *Nature* **463**, 899–905 (2010).
- Deshpande, A., Sicinski, P. & Hinds, P. W. Cyclins and cdk in development and cancer: a perspective. *Oncogene* **24**, 2909–2915 (2005).
- Musgrove, E. A., Caldon, C. E., Barraclough, J., Stone, A. & Sutherland, R. L. Cyclin D as a therapeutic target in cancer. *Nat. Rev. Cancer* **11**, 558–572 (2011).
- Malumbres, M. & Barbacid, M. To cycle or not to cycle: a critical decision in cancer. *Nat. Rev. Cancer* **1**, 222–231 (2001).
- Bertoli, C., Skotheim, J. M. & de Bruin, R. A. Control of cell cycle transcription during G1 and S phases. *Nat. Rev. Mol. Cell Biol.* **14**, 518–528 (2013).
- Matsuura, I. *et al.* Cyclin-dependent kinases regulate the antiproliferative function of Smads. *Nature* **430**, 226–231 (2004).
- Anders, L. *et al.* A systematic screen for CDK4/6 substrates links FOXM1 phosphorylation to senescence suppression in cancer cells. *Cancer Cell* **20**, 620–634 (2011).
- Aggarwal, P. *et al.* Nuclear cyclin D1/CDK4 kinase regulates CUL4 expression and triggers neoplastic growth via activation of the PRMT5 methyltransferase. *Cancer Cell* **18**, 329–340 (2010).
- Luscher-Firzlaff, J. M., Lilischkis, R. & Luscher, B. Regulation of the transcription factor FOXM1c by cyclin E/CDK2. *FEBS Lett.* **580**, 1716–1722 (2006).
- Hara, E., Hall, M. & Peters, G. Cdk2-dependent phosphorylation of Id2 modulates activity of E2A-related transcription factors. *EMBO J.* **16**, 332–342 (1997).
- Voit, R., Hoffmann, M. & Grummt, I. Phosphorylation by G1-specific cdk-cyclin complexes activates the nucleolar transcription factor UBF. *EMBO J.* **18**, 1891–1899 (1999).
- Yun, J. *et al.* Cdk2-dependent phosphorylation of the NF-Y transcription factor and its involvement in the p53-p21 signaling pathway. *J. Biol. Chem.* **278**, 36966–36972 (2003).
- Chae, H. D., Yun, J., Bang, Y. J. & Shin, D. Y. Cdk2-dependent phosphorylation of the NF-Y transcription factor is essential for the expression of the cell cycle-regulatory genes and cell cycle G1/S and G2/M transitions. *Oncogene* **23**, 4084–4088 (2004).
- Zhao, J. *et al.* NPAT links cyclin E-Cdk2 to the regulation of replication-dependent histone gene transcription. *Genes Dev.* **14**, 2283–2297 (2000).
- Ma, T. *et al.* Cell cycle-regulated phosphorylation of p220^{NPAT} by cyclin E/Cdk2 in Cajal bodies promotes histone gene transcription. *Genes Dev.* **14**, 2298–2313 (2000).
- Saville, M. K. & Watson, R. J. The cell-cycle regulated transcription factor B-Myb is phosphorylated by cyclin A/Cdk2 at sites that enhance its transactivation properties. *Oncogene* **17**, 2679–2689 (1998).
- Hydbring, P. *et al.* Phosphorylation by Cdk2 is required for Myc to repress Ras-induced senescence in cotransformation. *Proc. Natl Acad. Sci. USA* **107**, 58–63 (2010).
- Campaner, S. *et al.* Cdk2 suppresses cellular senescence induced by the *c-myc* oncogene. *Nat. Cell Biol.* **12**, 54–59 (2010).
- Rao, S. S., Chu, C. & Kohzt, D. S. Ectopic expression of cyclin D1 prevents activation of gene transcription by myogenic basic helix-loop-helix regulators. *Mol. Cell Biol.* **14**, 5259–5267 (1994).
- Skaepke, S. X., Rhee, J., Spicer, D. B. & Lassar, A. B. Inhibition of myogenic differentiation in proliferating myoblasts by cyclin D1-dependent kinase. *Science* **267**, 1022–1024 (1995).
- Skaepke, S. X., Rhee, J., Kim, P. S., Novitch, B. G. & Lassar, A. B. Cyclin-mediated inhibition of muscle gene expression via a mechanism that is independent of pRB hyperphosphorylation. *Mol. Cell Biol.* **16**, 7043–7053 (1996).
- Zhang, J. M., Wei, Q., Zhao, X. & Paterson, B. M. Coupling of the cell cycle and myogenesis through the cyclin D1-dependent interaction of MyoD with cdk4. *EMBO J.* **18**, 926–933 (1999).
- Shao, Z. & Robbins, P. D. Differential regulation of E2F and Sp1-mediated transcription by G1 cyclins. *Oncogene* **10**, 221–228 (1995).
- Adnane, J., Shao, Z. & Robbins, P. D. Cyclin D1 associates with the TBP-associated factor TAF_{II}250 to regulate Sp1-mediated transcription. *Oncogene* **18**, 239–247 (1999).
- Zwijnen, R. M. *et al.* CDK-independent activation of estrogen receptor by cyclin D1. *Cell* **88**, 405–415 (1997).
- Neuman, E. *et al.* Cyclin D1 stimulation of estrogen receptor transcriptional activity independent of cdk4. *Mol. Cell Biol.* **17**, 5338–5347 (1997).
- McMahon, C., Suthiphongchai, T., DiRenzo, J. & Ewen, M. E. P/CAF associates with cyclin D1 and potentiates its activation of the estrogen receptor. *Proc. Natl Acad. Sci. USA* **96**, 5382–5387 (1999).
- References 30–32 report a CDK-independent role of cyclin D1 in regulating the transcriptional activity of the oestrogen receptor, and an interaction between cyclin D1 and histone acetyltransferases.**
- Knudsen, K. E., Cavenee, W. K. & Arden, K. C. D-type cyclins complex with the androgen receptor and inhibit its transcriptional transactivation ability. *Cancer Res.* **59**, 2297–2301 (1999).
- Petre, C. E., Wetherill, Y. B., Daniels, M. & Knudsen, K. E. Cyclin D1: mechanism and consequence of androgen receptor co-repressor activity. *J. Biol. Chem.* **277**, 2207–2215 (2002).
- Petre-Draviam, C. E. *et al.* Specificity of cyclin D1 for androgen receptor regulation. *Cancer Res.* **63**, 4903–4913 (2003).
- Reutens, A. T. *et al.* Cyclin D1 binds the androgen receptor and regulates hormone-dependent signaling in a p300/CBP-associated factor (P/CAF)-dependent manner. *Mol. Endocrinol.* **15**, 797–811 (2001).
- Fu, M. *et al.* Cyclin D1 represses p300 transactivation through a cyclin-dependent kinase-independent mechanism. *J. Biol. Chem.* **280**, 29728–29742 (2005).
- Fu, M. *et al.* Cyclin D1 inhibits peroxisome proliferator-activated receptor γ -mediated adipogenesis through histone deacetylase recruitment. *J. Biol. Chem.* **280**, 16934–16941 (2005).
- Ratneau, C., Petry, M. W., Mutoh, H. & Leiter, A. B. Cyclin D1 represses the basic helix-loop-helix transcription factor, BETA2/NeuroD. *J. Biol. Chem.* **277**, 8847–8853 (2002).
- Iwatani, K., Fujimoto, T. & Ito, T. Cyclin D1 blocks the anti-proliferative function of RUNX3 by interfering with RUNX3–p300 interaction. *Biochem. Biophys. Res. Commun.* **400**, 426–431 (2010).
- Lin, H. M., Zhao, L. & Cheng, S. Y. Cyclin D1 is a ligand-independent co-repressor for thyroid hormone receptors. *J. Biol. Chem.* **277**, 28733–28741 (2002).
- Hirai, H. & Sherr, C. J. Interaction of D-type cyclins with a novel myb-like transcription factor, DMP1. *Mol. Cell Biol.* **16**, 6457–6467 (1996).
- Inoue, K. & Sherr, C. J. Gene expression and cell cycle arrest mediated by transcription factor DMP1 is antagonized by D-type cyclins through a cyclin-dependent-kinase-independent mechanism. *Mol. Cell Biol.* **18**, 1590–1600 (1998).
- Jian, Y. *et al.* Cyclin D3 interacts with vitamin D receptor and regulates its transcription activity. *Biochem. Biophys. Res. Commun.* **335**, 739–748 (2005).
- Liu, W. *et al.* Cyclin D3 interacts with human activating transcription factor 5 and potentiates its transcription activity. *Biochem. Biophys. Res. Commun.* **321**, 954–960 (2004).
- Peterson, L. F. *et al.* The hematopoietic transcription factor AML1 (RUNX1) is negatively regulated by the cell cycle protein cyclin D3. *Mol. Cell Biol.* **25**, 10205–10219 (2005).
- Lamb, J. *et al.* A mechanism of cyclin D1 action encoded in the patterns of gene expression in human cancer. *Cell* **114**, 323–334 (2003).
- This report revealed the relevance of CEBP β in the regulation of transcription by cyclin D1, using genome-wide analysis of tumour cells.**
- Bienvenu, F. *et al.* Transcriptional role of cyclin D1 in development revealed by a genetic-proteomic screen. *Nature* **463**, 374–378 (2010).
- This genetic-proteomic *in vivo* analysis revealed the transcriptional function of cyclin D1 during development.**
- Jadhav, A. P., Mason, H. A. & Cepko, C. L. Notch 1 inhibits photoreceptor production in the developing mammalian retina. *Development* **133**, 913–923 (2006).
- Sicinski, P. *et al.* Cyclin D1 provides a link between development and oncogenesis in the retina and breast. *Cell* **82**, 621–630 (1995).
- Fantl, V., Stamp, G., Andrews, A., Rosewell, I. & Dickson, C. Mice lacking cyclin D1 are small and show defects in eye and mammary gland development. *Genes Dev.* **9**, 2364–2372 (1995).
- Casimiro, M. C. *et al.* ChIP sequencing of cyclin D1 reveals a transcriptional role in chromosomal instability in mice. *J. Clin. Invest.* **122**, 833–843 (2012).
- Fujimoto, T., Anderson, K., Jacobsen, S. E., Nishikawa, S. I. & Nerlov, C. Cdk6 blocks myeloid differentiation by interfering with Runx1 DNA binding and Runx1-C/EBP α interaction. *EMBO J.* **26**, 2361–2370 (2007).

54. Kollmann, K. *et al.* A kinase-independent function of CDK6 links the cell cycle to tumor angiogenesis. *Cancer Cell* **24**, 167–181 (2013).
This study demonstrates a transcriptional, kinase-independent role of CDK6 in controlling cell proliferation and angiogenesis.
55. Yasui, M. *et al.* Antisense to cyclin D1 inhibits vascular endothelial growth factor-stimulated growth of vascular endothelial cells: implication of tumor vascularization. *Clin. Cancer Res.* **12**, 4720–4729 (2006).
56. Scheicher, R. *et al.* CDK6 as a key regulator of hematopoietic and leukemic stem cell activation. *Blood* **125**, 90–101 (2015).
57. Ira, G. *et al.* DNA end resection, homologous recombination and DNA damage checkpoint activation require CDK1. *Nature* **431**, 1011–1017 (2004).
58. Aylon, Y., Liefshitz, B. & Kupiec, M. The CDK regulates repair of double-strand breaks by homologous recombination during the cell cycle. *EMBO J.* **23**, 4868–4875 (2004).
59. Caspari, T., Murray, J. M. & Carr, A. M. Cdc2-cyclin B kinase activity links Crb2 and Rqh1-topoisomerase III. *Genes Dev.* **16**, 1195–1208 (2002).
60. Huertas, P., Cortes-Ledesma, F., Sartori, A. A., Aguilera, A. & Jackson, S. P. CDK targets Sae2 to control DNA-end resection and homologous recombination. *Nature* **455**, 689–692 (2008).
61. Saponaro, M. *et al.* Cdk1 targets Srs2 to complete synthesis-dependent strand annealing and to promote recombinational repair. *PLoS Genet.* **6**, e1000858 (2010).
62. Granata, M. *et al.* Dynamics of Rad9 chromatin binding and checkpoint function are mediated by its dimerization and are cell cycle-regulated by CDK1 activity. *PLoS Genet.* **6**, e1001047 (2010).
63. Wang, G., Tong, X., Weng, S. & Zhou, H. Multiple phosphorylation of Rad9 by CDK is required for DNA damage checkpoint activation. *Cycl. Cycle* **11**, 3792–3800 (2012).
64. Abreu, C. M. *et al.* Site-specific phosphorylation of the DNA damage response mediator Rad9 by cyclin-dependent kinases regulates activation of checkpoint kinase 1. *PLoS Genet.* **9**, e1003310 (2013).
65. Chen, X. *et al.* Cell cycle regulation of DNA double-strand break end resection by Cdk1-dependent Dna2 phosphorylation. *Nat. Struct. Mol. Biol.* **18**, 1015–1019 (2011).
66. Ferretti, L. P., Lafranchi, L. & Sartori, A. A. Controlling DNA-end resection: a new task for CDKs. *Front. Genet.* **4**, 99 (2013).
67. Esashi, F. *et al.* CDK-dependent phosphorylation of BRCA2 as a regulatory mechanism for recombinational repair. *Nature* **434**, 598–604 (2005).
This article reports that CDK-dependent phosphorylation controls the formation of BRCA2–RAD51 complexes.
68. Ayoub, N. *et al.* The carboxyl terminus of Brca2 links the disassembly of Rad51 complexes to mitotic entry. *Curr. Biol.* **19**, 1075–1085 (2009).
69. Yu, X., Wu, L. C., Bowcock, A. M., Aronheim, A. & Baer, R. The C-terminal (BRCT) domains of BRCA1 interact *in vivo* with CtIP, a protein implicated in the CtBP pathway of transcriptional repression. *J. Biol. Chem.* **273**, 25388–25392 (1998).
70. Agami, R. & Bernards, R. Distinct initiation and maintenance mechanisms cooperate to induce G1 cell cycle arrest in response to DNA damage. *Cell* **102**, 55–66 (2000).
71. Aggarwal, P. *et al.* Nuclear accumulation of cyclin D1 during S phase inhibits Cdk4-dependent Cdt1 proteolysis and triggers p53-dependent DNA rereplication. *Genes Dev.* **21**, 2908–2922 (2007).
72. Li, Z. *et al.* Alternative cyclin D1 splice forms differentially regulate the DNA damage response. *Cancer Res.* **70**, 8802–8811 (2010).
73. Jirawatnotai, S. *et al.* A function for cyclin D1 in DNA repair uncovered by protein interactome analyses in human cancers. *Nature* **474**, 230–234 (2011).
This proteome-wide analysis of cyclin D1-interacting proteins in human cancer cells reveals a role for cyclin D1 in recruiting RAD51 to DNA damage sites and in promoting homologous recombination.
74. Coco Martin, J. M., Balkenende, A., Verschoor, T., Lallemand, F. & Michalides, R. Cyclin D1 overexpression enhances radiation-induced apoptosis and radiosensitivity in a breast tumor cell line. *Cancer Res.* **59**, 1134–1140 (1999).
75. Shintani, S., Mihara, M., Ueyama, Y., Matsumura, T. & Wong, D. T. Cyclin D1 overexpression associates with radiosensitivity in oral squamous cell carcinoma. *Int. J. Cancer* **96**, 159–165 (2001).
76. Indovina, P., Pentimalli, F., Casini, N., Vocca, I. & Giordano, A. RB1 dual role in proliferation and apoptosis: cell fate control and implications for cancer therapy. *Oncotarget* **6**, 17873–17890 (2015).
77. Kranenburg, O., van der Eb, A. J. & Zantema, A. Cyclin D1 is an essential mediator of apoptotic neuronal cell death. *EMBO J.* **15**, 46–54 (1996).
78. Duquesne, F., Florent, M., Roue, G., Troussard, X. & Sola, B. Ectopic expression of cyclin D1 impairs the proliferation and enhances the apoptosis of a murine lymphoid cell line. *Cell Death Differ.* **8**, 51–62 (2001).
79. Roue, G., Pichereau, V., Lincet, H., Colomer, D. & Sola, B. Cyclin D1 mediates resistance to apoptosis through upregulation of molecular chaperones and consequent redistribution of cell death regulators. *Oncogene* **27**, 4909–4920 (2008).
80. Han, E. K. *et al.* Increased expression of cyclin D1 in a murine mammary epithelial cell line induces p27^{Kip1}, inhibits growth, and enhances apoptosis. *Cell Growth Differ.* **7**, 699–710 (1996).
81. Niu, M. Y., Menard, M., Reed, J. C., Krajewski, S. & Pratt, M. A. Ectopic expression of cyclin D1 amplifies a retinoic acid-induced mitochondrial death pathway in breast cancer cells. *Oncogene* **20**, 3506–3518 (2001).
82. Hochhauser, D. *et al.* Effect of cyclin D1 overexpression on drug sensitivity in a human fibrosarcoma cell line. *J. Natl. Cancer Inst.* **88**, 1269–1275 (1996).
83. Kornmann, M. *et al.* Inhibition of cyclin D1 expression in human pancreatic cancer cells is associated with increased chemosensitivity and decreased expression of multiple chemoresistance genes. *Cancer Res.* **59**, 3505–3511 (1999).
84. Boonen, G. J. *et al.* Cyclin D3 regulates proliferation and apoptosis of leukemic T cell lines. *J. Biol. Chem.* **274**, 34676–34682 (1999).
85. Choi, Y. J. *et al.* The requirement for cyclin D function in tumor maintenance. *Cancer Cell* **22**, 438–451 (2012).
86. Beltrán, E. *et al.* A cyclin-D1 interaction with BAX underlies its oncogenic role and potential as a therapeutic target in mantle cell lymphoma. *Proc. Natl. Acad. Sci. USA* **108**, 12461–12466 (2011).
87. Suzuki, A. *et al.* Survivin initiates procaspase 3/p21 complex formation as a result of interaction with Cdk4 to resist Fas-mediated cell death. *Oncogene* **19**, 1346–1353 (2000).
88. Mendelsohn, A. R., Hamer, J. D., Wang, Z. B. & Brent, R. Cyclin D3 activates Caspase 2, connecting cell proliferation with cell death. *Proc. Natl. Acad. Sci. USA* **99**, 6871–6876 (2002).
89. Osuga, H. *et al.* Cyclin-dependent kinases as a therapeutic target for stroke. *Proc. Natl. Acad. Sci. USA* **97**, 10254–10259 (2000).
90. osulno, H. & Chiba, T. Cyclin-dependent kinase 4 and cyclin D1 are required for excitotoxin-induced neuronal cell death *in vivo*. *J. Neurosci.* **21**, 6086–6094 (2001).
91. Ma, C., Papermaster, D. & Cepko, C. L. A unique pattern of photoreceptor degeneration in cyclin D1 mutant mice. *Proc. Natl. Acad. Sci. USA* **95**, 9938–9943 (1998).
92. Choi, Y. J. *et al.* D-cyclins repress apoptosis in hematopoietic cells by controlling death receptor Fas and its ligand FasL. *Dev. Cell* **30**, 255–267 (2014).
This study demonstrates the overlapping roles of D-type cyclins in controlling FAS-mediated apoptosis in haematopoietic cells.
93. Huang, H., Regan, K. M., Lou, Z., Chen, J. & Tindall, D. J. CDK2-dependent phosphorylation of FOXO1 as an apoptotic response to DNA damage. *Science* **314**, 294–297 (2006).
94. Choudhary, G. S. *et al.* Cyclin E/Cdk2-dependent phosphorylation of Mcl-1 determines its stability and cellular sensitivity to BH3 mimetics. *Oncotarget* **6**, 16912–16925 (2015).
95. Brown, N. E. *et al.* Cyclin D1 activity regulates autophagy and senescence in the mammary epithelium. *Cancer Res.* **72**, 6477–6489 (2012).
96. Jiang, H. *et al.* The RB-E2F1 pathway regulates autophagy. *Cancer Res.* **70**, 7882–7893 (2010).
97. Gan, L. *et al.* Cyclin D1 promotes anchorage-independent cell survival by inhibiting FOXO-mediated anoikis. *Cell Death Differ.* **16**, 1408–1417 (2009).
98. Shen, R. *et al.* Cyclin D1-Cdk4 induce Runx2 ubiquitination and degradation. *J. Biol. Chem.* **281**, 16347–16353 (2006).
99. Chen, S. *et al.* Cyclin-dependent kinases regulate epigenetic gene silencing through phosphorylation of EZH2. *Nat. Cell Biol.* **12**, 1108–1114 (2010).
100. Kaneko, S. *et al.* Phosphorylation of the PRC2 component Ezh2 is cell cycle-regulated and up-regulates its binding to ncRNA. *Genes Dev.* **24**, 2615–2620 (2010).
Together with reference 99, this study reports a link between CDKs and the epigenetic control of gene expression through the phosphorylation of PRCs.
101. Rao, S. S. & Kohtz, D. S. Positive and negative regulation of D-type cyclin expression in skeletal myoblasts by basic fibroblast growth factor and transforming growth factor. A role for cyclin D1 in control of myoblast differentiation. *J. Biol. Chem.* **270**, 4093–4100 (1995).
102. Lazaro, J. B., Bailey, P. J. & Lassar, A. B. Cyclin D–cdk4 activity modulates the subnuclear localization and interaction of MEF2 with SRC-family coactivators during skeletal muscle differentiation. *Genes Dev.* **16**, 1792–1805 (2002).
103. Nakajima, K. *et al.* Coordinated regulation of differentiation and proliferation of embryonic cardiomyocytes by a jumoni (Jarid2)-cyclin D1 pathway. *Development* **138**, 1771–1782 (2011).
104. Kiess, M., Gill, R. M. & Hamel, P. A. Expression of the positive regulator of cell cycle progression, cyclin D3, is induced during differentiation of myoblasts into quiescent myotubes. *Oncogene* **10**, 159–166 (1995).
105. Cenciarelli, C. *et al.* Critical role played by cyclin D3 in the MyoD-mediated arrest of cell cycle during myoblast differentiation. *Mol. Cell. Biol.* **19**, 5203–5217 (1999).
106. De Santa, F. *et al.* pRB-dependent cyclin D3 protein stabilization is required for myogenic differentiation. *Mol. Cell. Biol.* **27**, 7248–7265 (2007).
107. Bartkova, J., Lukas, J., Strauss, M. & Bartek, J. Cyclin D3: requirement for G1/S transition and high abundance in quiescent tissues suggest a dual role in proliferation and differentiation. *Oncogene* **17**, 1027–1037 (1998).
108. Dogliani, C. *et al.* Cyclin D3 expression in normal, reactive and neoplastic tissues. *J. Pathol.* **185**, 159–166 (1998).
109. Sarruf, D. A. *et al.* Cyclin D3 promotes adipogenesis through activation of peroxisome proliferator-activated receptor γ . *Mol. Cell. Biol.* **25**, 9985–9995 (2005).
110. Saavedra-Avila, N. A. *et al.* Cyclin D3 promotes pancreatic β -cell fitness and viability in a cell cycle-independent manner and is targeted in autoimmune diabetes. *Proc. Natl. Acad. Sci. USA* **111**, E3405–E3414 (2014).
111. Scinska, E. *et al.* Requirement for cyclin D3 in lymphocyte development and T cell leukemias. *Cancer Cell* **4**, 451–461 (2003).
112. Ajioka, I. Coordination of proliferation and neuronal differentiation by the retinoblastoma protein family. *Dev. Growth Differ.* **56**, 324–334 (2014).
113. Ali, F. *et al.* Cell cycle-regulated multi-site phosphorylation of Neurogenin 2 coordinates cell cycling with differentiation during neurogenesis. *Development* **138**, 4267–4277 (2011).
114. Hindley, C. *et al.* Post-translational modification of Ngn2 differentially affects transcription of distinct targets to regulate the balance between progenitor maintenance and differentiation. *Development* **139**, 1718–1723 (2012).
115. Lange, C., Huttner, W. B. & Calegari, F. Cdk4/cyclinD1 overexpression in neural stem cells shortens G1, delays neurogenesis, and promotes the generation and expansion of basal progenitors. *Cell Stem Cell* **5**, 320–331 (2009).
116. Lim, S. & Kaldis, P. Loss of Cdk2 and Cdk4 induces a switch from proliferation to differentiation in neural stem cells. *Stem Cells* **30**, 1509–1520 (2012).
117. Tamaru, T., Trigon, S. K., Okada, M. & Nakagawa, H. Identification of cells expressing a D type G1 cyclin in matured brain: implication for its role in neuronal function. *Neurosci. Lett.* **153**, 169–172 (1993).
118. Tamaru, T., Okada, M. & Nakagawa, H. Differential expression of D type cyclins during neuronal maturation. *Neurosci. Lett.* **168**, 229–232 (1994).
119. Ross, M. E., Carter, M. L. & Lee, J. H. MN20, a D2 cyclin, is transiently expressed in selected neural populations during embryogenesis. *J. Neurosci.* **16**, 210–219 (1996).
120. Lukaszewicz, A. I. & Anderson, D. J. Cyclin D1 promotes neurogenesis in the developing spinal cord in a cell cycle-independent manner. *Proc. Natl. Acad. Sci. USA* **108**, 11632–11637 (2011).

121. Malumbres, M. Physiological relevance of cell cycle kinases. *Physiol. Rev.* **91**, 975–1007 (2011).
122. Berger, C., Pallavi, S. K., Prasad, M., Shashidhara, L. S. & Technau, G. M. A critical role for cyclin E in cell fate determination in the central nervous system of *Drosophila melanogaster*. *Nat. Cell Biol.* **7**, 56–62 (2005).
123. Berger, C. *et al.* Cell cycle independent role of Cyclin E during neural cell fate specification in *Drosophila* is mediated by its regulation of Prospero function. *Dev. Biol.* **337**, 415–424 (2010).
124. Odajima, J. *et al.* Cyclin E constrains Cdk5 activity to regulate synaptic plasticity and memory formation. *Dev. Cell* **21**, 655–668 (2011).
This report describes an unexpected function of cyclin E in inhibiting CDK5 activity in terminally differentiated neurons, and in regulating synaptic plasticity, learning and memory.
125. Su, S. C. & Tsai, L. H. Cyclin-dependent kinases in brain development and disease. *Annu. Rev. Cell Dev. Biol.* **27**, 465–491 (2011).
126. Chunder, N., Wang, L., Chen, C., Hancock, W. W. & Wells, A. D. Cyclin-dependent kinase 2 controls peripheral immune tolerance. *J. Immunol.* **189**, 5659–5666 (2012).
127. Wells, A. D. & Morawski, P. A. New roles for cyclin-dependent kinases in T cell biology: linking cell division and differentiation. *Nat. Rev. Immunol.* **14**, 261–270 (2014).
128. Morawski, P. A., Mehra, P., Chen, C., Bhatti, T. & Wells, A. D. Foxp3 protein stability is regulated by cyclin-dependent kinase 2. *J. Biol. Chem.* **288**, 24494–24502 (2013).
129. Lee, J. & Desiderio, S. Cyclin A/CDK2 regulates V(D)J recombination by coordinating RAG-2 accumulation and DNA repair. *Immunity* **11**, 771–781 (1999).
130. Jiang, H., Ross, A. E. & Desiderio, S. Cell cycle-dependent accumulation *in vivo* of transposition-competent complexes between recombination signal ends and full-length RAG proteins. *J. Biol. Chem.* **279**, 8478–8486 (2004).
131. Fernandez-Hernandez, R. *et al.* Cyclin D1 localizes in the cytoplasm of keratinocytes during skin differentiation and regulates cell-matrix adhesion. *Cell Cycle* **12**, 2510–2517 (2013).
132. Bendris, N., Lemmers, B. & Blanchard, J. M. Cell cycle, cytoskeleton dynamics and beyond: the many functions of cyclins and CDK inhibitors. *Cell Cycle* **14**, 1786–1798 (2015).
133. Fahraeus, R. & Lane, D. P. The p16^{INK4a} tumour suppressor protein inhibits $\alpha_5\beta_3$ integrin-mediated cell spreading on vitronectin by blocking PKC-dependent localization of $\alpha_5\beta_3$ to focal contacts. *EMBO J.* **18**, 2106–2118 (1999).
134. Li, Z. *et al.* Cyclin D1 regulates cellular migration through the inhibition of thrombospondin 1 and ROCK signaling. *Mol. Cell. Biol.* **26**, 4240–4256 (2006).
135. Li, Z. *et al.* Cyclin D1 induction of cellular migration requires p27^{Kip1}. *Cancer Res.* **66**, 9986–9994 (2006).
136. Li, Z. *et al.* Alternate cyclin D1 mRNA splicing modulates p27^{Kip1} binding and cell migration. *J. Biol. Chem.* **283**, 7007–7015 (2008).
137. Zhong, Z. *et al.* Cyclin D1/cyclin-dependent kinase 4 interacts with filamin A and affects the migration and invasion potential of breast cancer cells. *Cancer Res.* **70**, 2105–2114 (2010).
138. Yu, Z. *et al.* Cyclin D1 induction of Dicer governs microRNA processing and expression in breast cancer. *Nat. Commun.* **4**, 2812 (2013).
This report describes the control of DICER by cyclin D1 and its various implications for understanding the interactions between cell cycle proteins and non-coding RNAs.
139. Lopez-Mejia, I. C. & Fajas, L. Cell cycle regulation of mitochondrial function. *Curr. Opin. Cell Biol.* **33**, 19–25 (2015).
140. Sakamaki, T. *et al.* Cyclin D1 determines mitochondrial function *in vivo*. *Mol. Cell. Biol.* **26**, 5449–5469 (2006).
141. Wang, C. *et al.* Cyclin D1 repression of nuclear respiratory factor 1 integrates nuclear DNA synthesis and mitochondrial function. *Proc. Natl Acad. Sci. USA* **103**, 11567–11572 (2006).
Together with reference 140, this study establishes a molecular link between cyclin D1 and control of mitochondrial function through the inhibition of NRF1.
142. Tchakarska, G., Roussel, M., Troussard, X. & Sola, B. Cyclin D1 inhibits mitochondrial activity in B cells. *Cancer Res.* **71**, 1690–1699 (2011).
143. Kaplon, J., van Dam, L. & Peeper, D. Two-way communication between the metabolic and cell cycle machineries: the molecular basis. *Cell Cycle* **14**, 2022–2032 (2015).
144. Xu, Y., Swartz, K. L., Siu, K. T., Bhattacharyya, M. & Minella, A. C. Fbw7-dependent cyclin E regulation ensures terminal maturation of bone marrow erythroid cells by restraining oxidative metabolism. *Oncogene* **33**, 3161–3171 (2014).
145. Wang, Z. *et al.* Cyclin B1/Cdk1 coordinates mitochondrial respiration for cell-cycle G2/M progression. *Dev. Cell* **29**, 217–232 (2014).
146. Rane, S. G. *et al.* Loss of Cdk4 expression causes insulin-deficient diabetes and Cdk4 activation results in β -islet cell hyperplasia. *Nat. Genet.* **22**, 44–52 (1999).
147. Tsutsui, T. *et al.* Targeted disruption of CDK4 delays cell cycle entry with enhanced p27^{Kip1} activity. *Mol. Cell. Biol.* **19**, 7011–7019 (1999).
148. Georgia, S. & Bhushan, A. β cell replication is the primary mechanism for maintaining postnatal β cell mass. *J. Clin. Invest.* **114**, 963–968 (2004).
149. Kushner, J. A. *et al.* Cyclins D2 and D1 are essential for postnatal pancreatic β -cell growth. *Mol. Cell. Biol.* **25**, 3752–3762 (2005).
150. Annicotte, J. S. *et al.* The CDK4–pRB–E2F1 pathway controls insulin secretion. *Nat. Cell Biol.* **11**, 1017–1023 (2009).
This study provides a molecular link between control of the cell cycle and insulin secretion in pancreatic β -cells.
151. Lee, Y. *et al.* Cyclin D1–Cdk4 controls glucose metabolism independently of cell cycle progression. *Nature* **510**, 547–551 (2014).
152. Hanse, E. A. *et al.* Cyclin D1 inhibits hepatic lipogenesis via repression of carbohydrate response element binding protein and hepatocyte nuclear factor 4 α . *Cell Cycle* **11**, 2681–2690 (2012).
153. Phelps, D. E. & Xiong, Y. Regulation of cyclin-dependent kinase 4 during adipogenesis involves switching of cyclin D subunits and concurrent binding of p18^{INK4c} and p27^{Kip1}. *Cell Growth Differ.* **9**, 595–610 (1998).
154. Abella, A. P. *et al.* Cdk4 promotes adipogenesis through PPAR γ activation. *Cell Metab.* **2**, 239–249 (2005).
This article reports that adipogenesis is induced by a direct interaction between cyclin D3–CDK4 and PPAR γ , a master regulator of adipogenesis.
155. Aguilar, V. & Fajas, L. Cycling through metabolism. *EMBO Mol. Med.* **2**, 338–348 (2010).
156. Turner, N. C., Huang Bartlett, C. & Cristofanilli, M. Palbociclib in hormone-receptor-positive advanced breast cancer. *N. Engl. J. Med.* **373**, 1672–1673 (2015).
157. Asghar, U., Witkiewicz, A. K., Turner, N. C. & Knudsen, E. S. The history and future of targeting cyclin-dependent kinases in cancer therapy. *Nat. Rev. Drug Discov.* **14**, 130–146 (2015).
158. Finn, R. S. *et al.* PD 0332991, a selective cyclin D kinase 4/6 inhibitor, preferentially inhibits proliferation of luminal estrogen receptor-positive human breast cancer cell lines *in vitro*. *Breast Cancer Res.* **11**, R77 (2009).
159. Sherr, C. J. & Roberts, J. M. CDK inhibitors: positive and negative regulators of G1-phase progression. *Genes Dev.* **13**, 1501–1512 (1999).
160. Gil, J. & Peters, G. Regulation of the *INK4b–ARF–INK4a* tumour suppressor locus: all for one or one for all. *Nat. Rev. Mol. Cell Biol.* **7**, 667–677 (2006).
161. Besson, A., Dowdy, S. F. & Roberts, J. M. CDK inhibitors: cell cycle regulators and beyond. *Dev. Cell* **14**, 159–169 (2008).
162. Caldon, C. E., Sutherland, R. L. & Musgrove, E. Cell cycle proteins in epithelial cell differentiation: implications for breast cancer. *Cell Cycle* **9**, 1918–1928 (2010).
163. Myatt, S. S. & Lam, E. W. Promiscuous and lineage-specific roles of cell cycle regulators in haematopoiesis. *Cell Div.* **2**, 6 (2007).
164. Beukelaers, P. *et al.* Cycling or not cycling: cell cycle regulatory molecules and adult neurogenesis. *Cell. Mol. Life Sci.* **69**, 1493–1503 (2012).
165. Coqueret, O. New roles for p21 and p27 cell-cycle inhibitors: a function for each cell compartment? *Trends Cell Biol.* **13**, 65–70 (2003).
166. Ceruti, J. M., Scassa, M. E., Flo, J. M., Varone, C. L. & Canepa, E. T. Induction of p19^{INK4d} in response to ultraviolet light improves DNA repair and confers resistance to apoptosis in neuroblastoma cells. *Oncogene* **24**, 4065–4080 (2005).
167. Ogara, M. F. *et al.* Chromatin relaxation-mediated induction of p19^{INK4d} increases the ability of cells to repair damaged DNA. *PLoS ONE* **8**, e61143 (2013).
168. Denicourt, C. & Dowdy, S. F. Cip/Kip proteins: more than just CDK inhibitors. *Genes Dev.* **18**, 851–855 (2004).
169. Nagahara, H. *et al.* Transduction of full-length TAT fusion proteins into mammalian cells: TAT-p27^{Kip1} induces cell migration. *Nat. Med.* **4**, 1449–1452 (1998).
170. McAllister, S. S., Becker-Hapak, M., Pintucci, G., Pagano, M. & Dowdy, S. F. Novel p27^{Kip1} C-terminal scatter domain mediates Rac-dependent cell migration independent of cell cycle arrest functions. *Mol. Cell. Biol.* **23**, 216–228 (2003).
171. Besson, A., Gurian-West, M., Schmidt, A., Hall, A. & Roberts, J. M. p27^{Kip1} modulates cell migration through the regulation of RhoA activation. *Genes Dev.* **18**, 862–876 (2004).
172. Liu, Q. *et al.* Cyclin D1 and C/EBP β LAP1 operate in a common pathway to promote mammary epithelial cell differentiation. *Mol. Cell. Biol.* **34**, 3168–3179 (2014).
173. Bienvenu, F., Gascan, H. & Coqueret, O. Cyclin D1 represses STAT3 activation through a Cdk4-independent mechanism. *J. Biol. Chem.* **276**, 16840–16847 (2001).

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Competing interests statement

The authors declare no competing interests.